IoT Security Research Project

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Overview

- Introduction to the Internet of Things
- Intro to Hardware and Software Systems
- IoT Vulnerability Research
  - Hypothesis
- Additional Research
- Takeaways from the project

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Introduction to the Internet of Things (IoT)

The Internet of Things (IoT)

- **IoT**
  - "There are 26.66 billion active IoT devices in 2019"(Nick G, Techjury)
  - EX. Coke vending machines, smart watches, Smart Home lights, and many more.

- **What is the internet of things?**
  - The IoT is a system of interconnected devices
  - IoT devices are used in many homes and office environments
    - Ex. Alexa and Google Home devices
  - Specific identifiers, UIDs, are used to guide direct data transfers between devices

- **IoT History**
  - IoT first started with a project done at the Carnegie Mellon university, where students got together to develop a basic system to keep track of how many cans a vending machine had.

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Intro Hardware and Software Systems

OSI Model
(Open Systems Interconnection)

- **OSI** : "an international standardization model to facilitate communications among computers with different protocols" (dictionary.com)
- The OSI model Consist of 7 Communication layers
  - Application, Presentation, Session, Transport, Network, Data Link, and Physical
- Each communication layer may have vulnerabilities
- Vulnerability analysis should take place at each layer

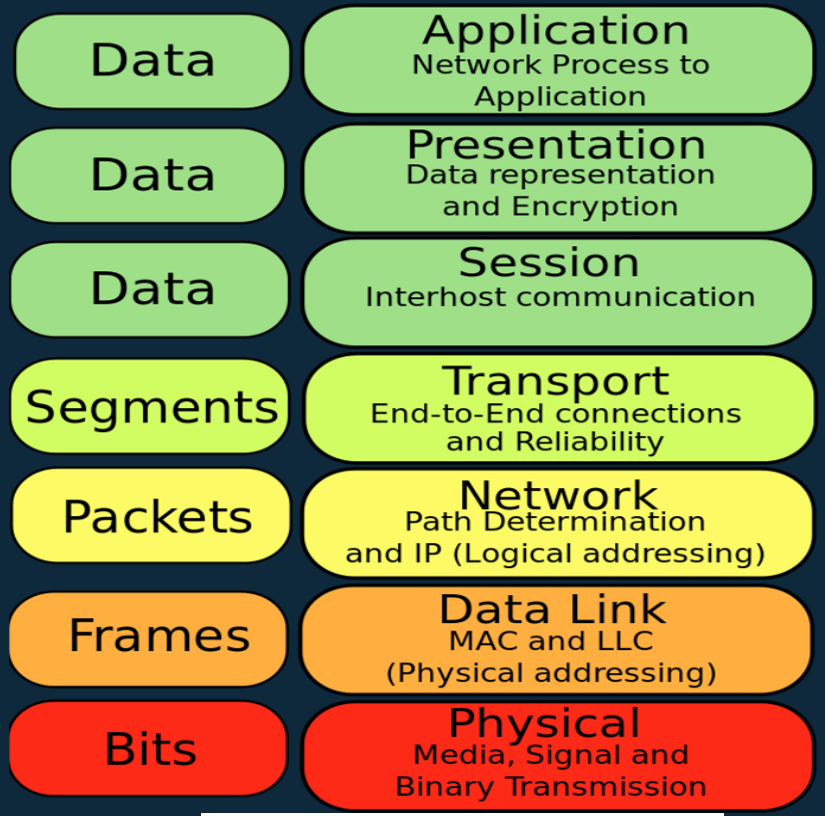


Image from Gorivero / Wikimedia Commons / CC-BY-SA-3.0

OSI Model
(Open Systems Interconnection)

Data	**Application** – Network Process to Application
Data	**Presentation** – Data representation and Encryption
Data	**Session** – Interhost communication
Segments	**Transport** – End-to-End connections and Reliability
Packets	**Network** – Path Determination and IP (Logical addressing)
Frames	**Data Link** – MAC and LLC (Physical addressing)
Bits	**Physical** – Media, Signal and Binary Transmission

- Browser places packets together in a webpage
- Breaking down and transporting the packages
- Ip addresses send packets between devices
- Electrical signals

Image from Gorivero / Wikimedia Commons / CC-BY-SA-3.0

UART Data

(Universal Asynchronous Receiver-Transmitter)

- UART is communication between the chips on the board and an external device
  - Using a controlled board such as a raspberry pi communications can be interrupted
- 4 Base UART pins
  - Gnd (ground connection)
  - Data output: receive (RX)
  - Data input: transmit (TX)
  - Voltage (vcc)
- UART communication can be exploited to control a IoT device

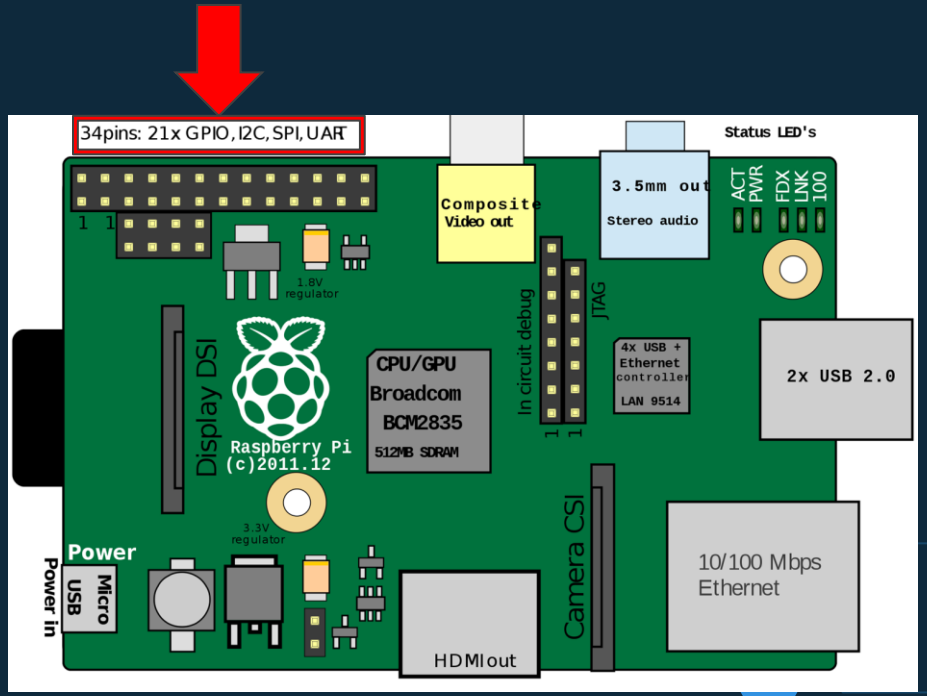


Image retrieved from https://www.quinaptis.com/2019/03/raspberrypi/

JTAG

JTAG was named after the Joint Test Action Group developed to debug chips

- Similar to UART data, JTAG data can be used to exploit an IoT device
- TAP (Test Access Port) is the name given to the JTAG interfaces.

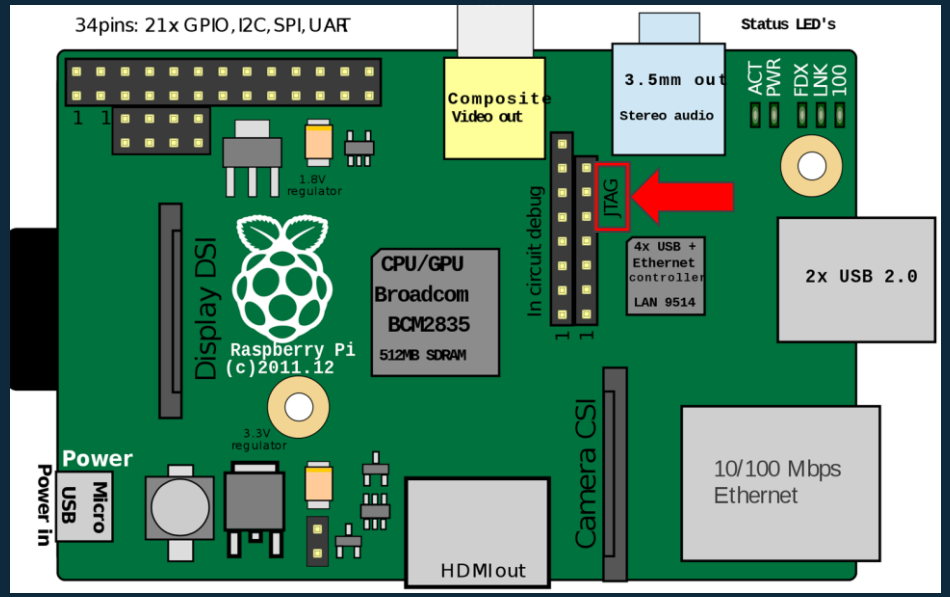


Image retrieved from https://www.quinaptis.com/2019/03/raspberrypi/

FCC (The Federal Communication Commision)

- The FCC is a government institution that monitors and manages radio and radio-like communications between devices
- The FCC ID is a tag that most radio devices have. This ID refers to a index where the FCC provides information about the communications used in the device.
- In the world of cybersecurity the information provided allows analyst to see the frequency and communication protocol used by the devices
- The information then allows for people to search on certain wavelengths to fetch data

Operating Frequencies

Frequency Range	Power Output	Rule Parts	Line Entry
2.412-2.462 GHz	291 mW	15C	1

(FCCID.io)

DATA Sniffing and Wiretapping Legalities

- Software Defined Radio (SDR) antennas can be used to capture data packets from devices
- Signals that have not been encrypted correctly and that are insecure forms of communication can leak information
- Laws addressing wiretapping place repercussions for listening to data that does not belong to the host

Image Retrieved from https://ezcontents.org/software-defined-radio-rtl-sdr-usb-dongle

3

IoT Vulnerability Research

Mirai Botnet

- In the past, Malware attacks have shown insecurities in IoT devices
- The Mirai Botnet Attack was a malware driven compromise of the internet of things
  - The malware used widespread DDoS attacks in order to compromise devices
  - Each additional device compromised then searched for more vulnerable devices to add to the expanding botnet
- Mirai took advantage of insecure default passwords.

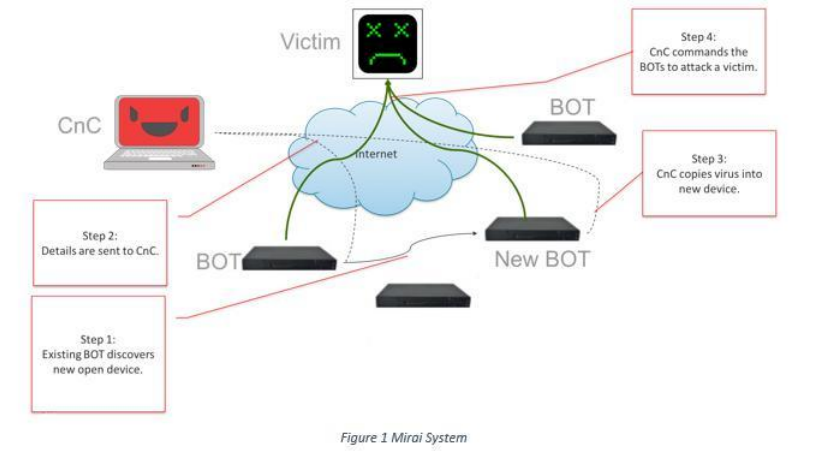


Figure 1 Mirai System

- **<u>The Impact:</u> After Mirai the internet released the source code for the attack causing security researchers to be concerned for future attacks of a similar nature that may occur**

Image from https://opensourceforu.com/2019/10/the-role-of-open-source-tools-and-concepts-in-iot-security/

Technical Vulnerabilities

(common methods for device infiltration)

- **Default user IDs and passwords**
  - **Device is insecure**
  - **By using the default id and password unauthorized users can connect to and infiltrate the device**
  - **This method was key in the mirai botnet**
- **IP Address and Network Vulnerabilities**
  - **By leaks in IP addresses and insecure networks exploiters can gain access to user interfaces and control a device or deny service to the device**

Technical Vulnerabilities (Cont.)

- Insecure Data Transfer
  - Through basic internet data sniffing and network monitoring packets can be leaked
  - If data is not secure and does not have safe encryption data can be stolen
- Insecure Access Points
  - Ex: insecure mobile applications
  - Using testing to gain access to insecure points in application systems can leak data and give total access to the control systems

Hypothesis

If these patterns in device vulnerabilities are repeated in the future, Smart Home and Office spaces will be at risk.

Developing the Alexa Farm

- **To start the Project I helped members in the lab setup devices in an attempt to demonstrate a possible Smart Office environment with integrated alexa devices**
- **This setup was connected to a internet proxy with a raspberry pi in order to be able to monitor data capture within the lab environment and record it**

1. Amazon Echo (alexa)
2. Lightbulb
3. Printer
4. TV
5. Air data collection device
6. Camera
7. Robot Vacuum
8. Thermostat
9. A lock

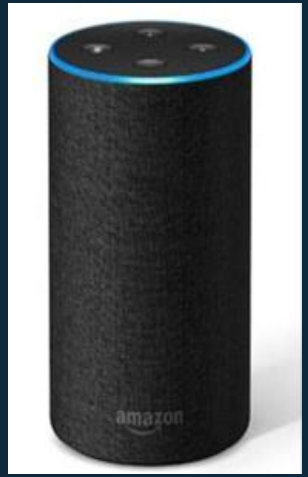


Image from https://www.zdnet.com/pictures/internet-of-things-gadgets-to-make-your-home-smarter/3/

Creating data capture for the office environment

- In the lab two data collection systems were studied.
- Both of these softwares collect records of all the data packets sent within the internet connection between devices.
- These records can be used for monitoring and recording the experiments and testing in the lab
- <u>ELK</u>
  - E = elasticsearch (Search engine)
  - L = logstash (Translating data)
  - K = kibana (user interface)
- <u>ROCK NSM</u>
  - Uses a ELK system with other softwares to better monitor internet connection

Image from https://rocknsm.io/

Rock NSM

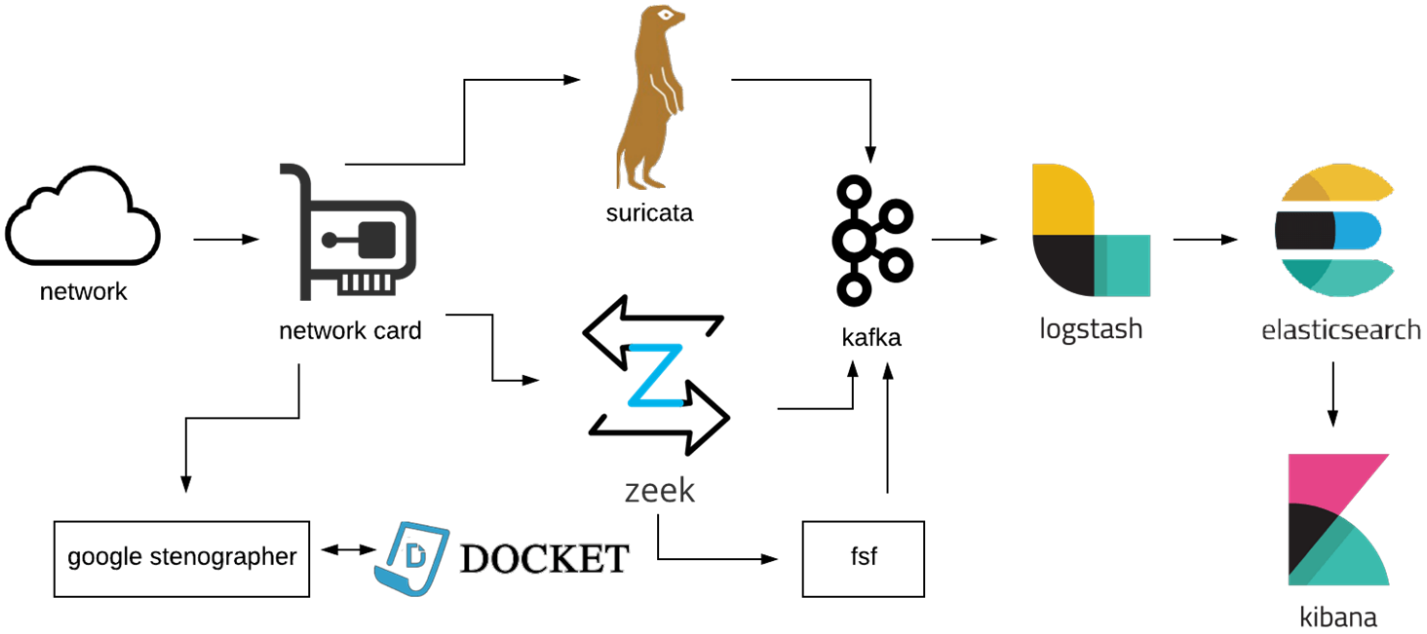
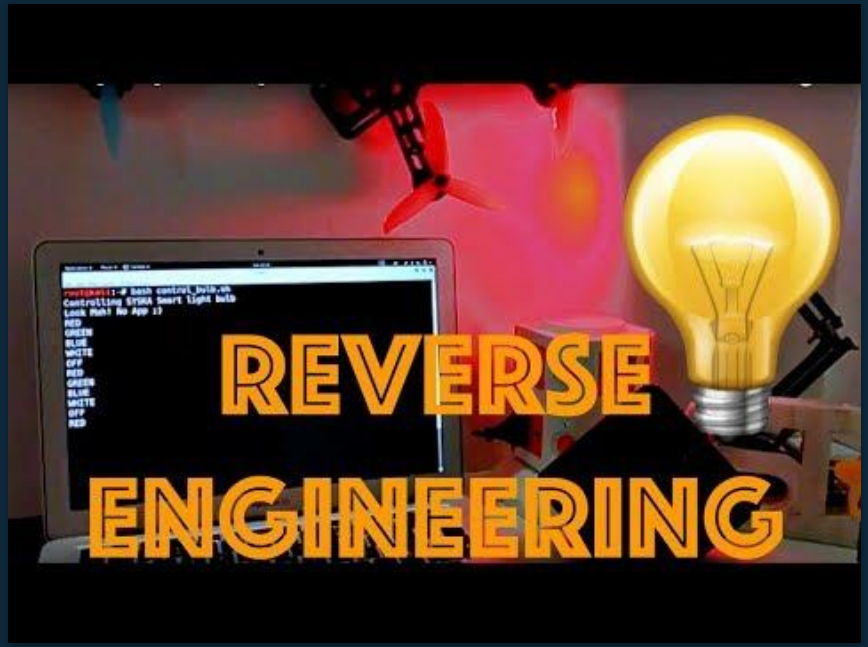


Image from https://rocknsm.io/

Testing Devices (Lightbulb)

- **Using Data packets captured, the lab was able to test responses in lightbulbs sending those packets to evoke unauthorized commands**



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Additional Research/Future Expectations

Lightbulb Network Data Breach

Shodan

Top Voted

11,906
Webcam
best ip cam search I have found yet.
webcam surveillance cams 2010-03-15

4,918
Cams
admin admin
cam webcam 2012-02-06

2,560
Netcam
Netcam
netcam 2012-01-13

1,947
default password
Finds results with "default password" in the ba...

Recently Shared

2
India webcams
webcams of India
webcam ind-cam camera 2020-05-

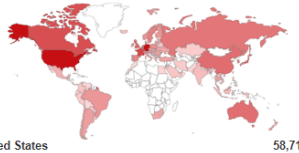
1
ICTV
2020-05-

1
embed
httpd!@#dvdndicidcndinscpdincspidncdpsincsndc
2020-05-

1
logitech

TOTAL RESULTS
203,501

TOP COUNTRIES



United States	58,714
Germany	56,366
France	16,472
Canada	12,424
United Kingdom	7,203

TOP ORGANIZATIONS

OVH SAS	22,465
marbis GmbH	19,776
Comcast Cable	8,360
Link11 GmbH	8,248
Spectrum	6,116

TOP OPERATING SYSTEMS

Linux 3.x	14
Linux 2.6.x	3
Windows XP	1
Windows Server 2008	1
Windows 7 or 8	1

TOP PRODUCTS

Minecraft	203,501

New Service: Keep track of what you ha

RELATED TAGS: new lab linux

[redacted] ppp.infoweb.ne.jp
Fujitsu
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5 Takeaways

Things to remember

- Not all devices are safe! Be sure to keep updated!
- IoT researchers are continuing to learn more and test vulnerabilities
- Data sniffing is illegal! Still watch out!
- Possible Mitigations can solve insecurities in IoT

Special Thanks to

- Oren Upton
- Julia Cottle
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- David King
- UTSA
- BASIS
- Teachers

Sources

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